Raspberry Pi Pico

P1
rpi pico 1 W
2648-SC0918CT-ND
smd

+3.3V
R1 220
3V1 green
GND

R10 4.7k
GND

+BATT
+3.3V
5V
J1

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

bt1 bt2 bt3 bt4

D1 D_Schottky
5V
+3.3V

R2 220
D13 blue
GND

sda
scl
right_2
right_1
right_en
left_2
left_en
left_1

J3 joystick
J4 speed knob

VBUS VSYS 3V3_OUT ADC_VREF
3V3_EN RUN
GP0 GP1 GP2 GP3 GP4 GP5 GP6 GP7 GP8 GP9 GP10 GP11 GP12
GP13 GP14 GP15 GP16 GP17 GP18 GP19 GP20 GP21 GP22 GP26 GP27 GP28
AGND GND GND_1 GND_2 GND_3 GND_4 GND_5 GND_6

* these pins are 5V tolerant inputs.
The input is divided down from 5v to a safe level.
If you want to use this pin as a 3.3v input or as an output,
solder the NetTie pads on the back of the board.

left motor

U1
VNH7040AYTR

HSL1
Heatsink

J100 0197054001
J101 0197054001
optionally replace J7 to be weelye compatible
J7 1017491

ML+ ML-

C1 0.1uF

left_sense
left_en
left_2

right motor

U2
VNH7040AYTR

C2 0.1uF

right_1
right_en
right_sense

J8 1017491
optionally replace J8 to make the PCB weelye compatible
weelye style J104
weelye style J105
J106 0197054001
J107 0197054001

HSR1
Heatsink

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR

+3.3V NT1
NetTie_3
TP0 TestPoint
+BATT
R13 100k
R14 8.2k
right_ipropi
left_ipropi
ADDR AIN0 AIN1 AIN2 AIN3 NC VDD SCL SDA GND
scl
sda
C4 0.1uF
pullup resistors
left_sense
right_sense
R17 220
R18 220
R11 442
R12 442
D6 3V zener
R15 3.3k
R16 3.3k

qwiic I2C plug
J2 Conn_01x04

weelye compatible header
J11 Conn_01x10_Pin
switch
bat-
D2 D_Schottky

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1 B2 B3 B4
ASJ-200-B-HT

battery power circuit
J102 0197054001
J103 0197054001
optionally replace J5 to be weelye compatible
J5 TB005-762-02BE
screw terminal
SW1
switch
R3 3.3k
R4 22.1k
R5 22.1k
18V zener
D4
D3
PSMN1R0-40YLDX
Q1 on/off switch
Q2 reverse polarity protection
C5 33uf
J6

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=49A4BE669EF8F291
C6 0.1uF
U3
LMR54410DBVR
L1 10uH
C3 1uF
C7 0.1uF
R8 115k
R9 22.1k
C8 22uF

optional cooling fans
wire to J6
MF40200V1-1000U-A99 (5V, 3.5-6)
MF40201V1-1000U-A99 (12V, 4.5-13.8)
MF40202V1-1000U-A99 (24V, 8-27.6V)
M1
M2

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com

Sheet: /
File: gbg-pcb.kicad_sch

Title: GBG-PCB

Size: A3	Date:	Rev: v2.4
KiCad E.D.A. 10.0.7		Id: 1/1